

1. B)

Objective: Find which of the QEC codes can correct for a single spontaneous emission error using the atomic qubit.

QEC codes described in Lec 2:

1. 3-qubit X repetition code: protects against Z-errors
2. 3-qubit Z repetition code: protects against X-errors / phase flip errors
3. 9-qubit Shor's code: protects against X & Z errors.

Now, for a QEC code to be valid, it must satisfy the KL conditions:

$$P E_a^\dagger E_b P = \alpha_{ab} P$$

$$P|\psi_L\rangle = |\psi_L\rangle \quad ; \quad \langle\psi_L|P = \langle\psi_L|$$

$$\begin{aligned} \langle\psi_L|P E_a^\dagger E_b P|\psi_L\rangle &= \alpha_{ab} \langle\psi_L|P|\psi_L\rangle \\ \langle\psi_L|E_a^\dagger E_b|\psi_L\rangle &= \alpha_{ab} \end{aligned}$$

$$\underline{\underline{\alpha_{ab}}} = \langle \psi_L | E_a^\dagger E_b | \psi_L \rangle$$

this should be the same for $|0_L\rangle$ and $|1_L\rangle$

Let's consider the 3-bit repetition code for bit-flips

$$|\psi_L\rangle \in \{|0_L\rangle, |1_L\rangle\}$$

$$|0_L\rangle = |000\rangle \quad ; \quad |1_L\rangle = |111\rangle$$

$$K_1, K_2 \in \{E_a\}^2$$

$$K_1 = |0\rangle\langle 0| + \sqrt{1-p} |1\rangle\langle 1| \quad ; \quad K_2 = \sqrt{p} |0\rangle\langle 1|$$

Let's consider K_1 on qubit 1.

$$E = K_1 \otimes I \otimes I$$

$$\& E^\dagger E = K_1^\dagger K_1 \otimes I \otimes I$$

$$= (|0\rangle\langle 0| + (p-1)|1\rangle\langle 1|) \otimes I \otimes I$$

$$\langle 000 | (|0\rangle\langle 0| + (p-1)|1\rangle\langle 1| \otimes I \otimes I) | 000 \rangle = 1$$

$$\langle 111 | (|0\rangle\langle 0| + (p-1)|1\rangle\langle 1| \otimes I \otimes I) | 111 \rangle = p-1$$

$$1 \neq p-1$$

So, KL conditions fail.

Let's consider the 3-bit phase repetition code:

$$|0_L\rangle = \frac{|000\rangle + |011\rangle + |101\rangle + |110\rangle}{2}$$

$$|1_L\rangle = \frac{|100\rangle + |010\rangle + |001\rangle + |111\rangle}{2}$$

actually, let's consider $K_2 = \sqrt{P} |0\rangle\langle 1|$ because it's a simpler operator. $K_2^\dagger K_2 = P |1\rangle\langle 1|$

$$\begin{aligned} & \langle 0_L | K_2^\dagger K_2 \otimes I \otimes I | 0_L \rangle \\ &= \langle 0_L | P | 1 \rangle \langle 1 | \otimes I \otimes I | 0_L \rangle \\ &= P \left(\frac{\langle 000 | + \langle 011 | + \langle 101 | + \langle 110 |}{2} \right) \left(\frac{| 101 \rangle + | 110 \rangle}{2} \right) \\ &= \frac{P}{4} \begin{pmatrix} 1 & +1 \end{pmatrix} \quad \text{since these states are orthogonal.} \\ &= P/2. \end{aligned}$$

$$\begin{aligned} & \langle 1_L | P | 1 \rangle \langle 1 | \otimes I \otimes I | 1_L \rangle \\ &= P \left(\langle 1_L | \frac{| 1100 \rangle + | 1111 \rangle}{2} \right) \\ &= \frac{P}{4} \begin{pmatrix} 2 \end{pmatrix} = P/2. \end{aligned}$$

Since there is symmetry, K_2 acting on any of the 3 qubits would $= P/2$.

Let's check $K_1^\dagger K_1$

$$= |0\rangle\langle 0| + (P-1)|1\rangle\langle 1|$$

$$\langle 0_L | (|0\rangle\langle 0| + (P-1)|1\rangle\langle 1|) \otimes I \otimes I | 0_L \rangle$$

$$= \langle 0_L | \left(\frac{|000\rangle + |011\rangle}{2} + (P-1) \frac{|101\rangle + |110\rangle}{2} \right)$$

$$= \frac{\cancel{1+1}}{2} + (P-1) \frac{\cancel{1+1}}{2}$$

$$= P.$$

Similarly,

$$\langle 1_L | (|0\rangle\langle 0| + (P-1)|1\rangle\langle 1|) \otimes I \otimes I | 1_L \rangle$$

$$= \langle 1_L | \left[\left(\frac{|1010\rangle + |1001\rangle}{2} \right) + \frac{P-1}{2} \left(\frac{|1111\rangle + |1100\rangle}{2} \right) \right]$$

$$= \frac{\cancel{(1+1)}}{2} + \frac{P-1}{2} \frac{\cancel{(1+1)}}{2} = P$$

$$\begin{aligned}
& K_2^\dagger K_1 \\
&= \sqrt{p} |1\rangle \langle 0| (|0\rangle \langle 0| + \sqrt{p-1} |1\rangle \langle 1|) \\
&= \sqrt{p} |1\rangle \langle 0|
\end{aligned}$$

$$\begin{aligned}
& \langle 0_L | (\sqrt{p} |1\rangle \langle 0|) \otimes I \otimes I | 0_L \rangle \\
&= \sqrt{p} \langle 0_L | \left(\frac{|1100\rangle + |1111\rangle}{2} \right) \\
&= 0
\end{aligned}$$

Similarly,

$$\begin{aligned}
& \langle 1_L | (\sqrt{p} |1\rangle \langle 0|) \otimes I \otimes I | 1_L \rangle \\
&= \sqrt{p} \langle 1_L | \left(\frac{|1110\rangle + |1101\rangle}{2} \right) \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
& K_1^\dagger K_2 \\
&= (|0\rangle \langle 0| + \sqrt{p-1} |1\rangle \langle 1|) \sqrt{p} |0\rangle \langle 1| \\
&= \sqrt{p} |0\rangle \langle 1| \\
& \langle 0_L | (\sqrt{p} |0\rangle \langle 1| \otimes I \otimes I) | 0_L \rangle \\
&= \sqrt{p} \langle 0_L | \left(\frac{|1001\rangle + |1010\rangle}{2} \right) = 0
\end{aligned}$$

Similarly,

$$\begin{aligned} & \langle 1_L | (\sqrt{P} |0\rangle \langle 1| \otimes I \otimes I) | 1_L \rangle \\ &= \sqrt{P} \langle 1_L | \left(\frac{|1000\rangle + |0111\rangle}{2} \right) \\ &= 0. \end{aligned}$$

Therefore, 3-bit phase repetition code can protect against the spontaneous emission.

... OED.

Important intuition / insight:

for the bit-flip code, ^{1st qubit}

$$|0_L\rangle = |000\rangle \Rightarrow P(\text{emission}) = 0$$

$$|1_L\rangle = |111\rangle \Rightarrow P(\text{emission}) = P$$

the environment can distinguish them statistically,

photon emission more likely $|1_L\rangle$ than $|0_L\rangle$

dangerous because environment knows something about the system.

for the phase-flip code,

$$|0_L\rangle = \frac{|000\rangle + |011\rangle + |101\rangle + |110\rangle}{2}$$

$$1^{\text{st}} \text{ qubit} \Rightarrow P(\text{emission}) = \frac{2P}{2} = P.$$

$$|1_L\rangle = \frac{|111\rangle + |100\rangle + |001\rangle + |010\rangle}{2}$$

$$1^{\text{st}} \text{ qubit} \Rightarrow P(\text{emission}) = \frac{2P}{2} = P.$$

$\therefore$ the environment can't distinguish them statistically.